

Full Stack Application Documentation

Adapted for a ServiceNow full-stack solution (Service Portal frontend, Flow Designer backend, custom table database)

Date	3 October 2026
Team ID	
Project Title	Automated Network Request Management System

1. Introduction

Project Title	Automated Network Request Management System
Team Members	To be added by the team (names and roles).

2. Project Overview

Purpose: Replace manual, email-based network requests with a standard, validated and automated process. Employees request network access, Wi-Fi, VPN or static IP; managers approve; IT receives complete records; everyone receives automated status emails.

Features:

- Network Request catalog item in the Service Portal with a Requester Information variable set.
- Dynamic form validation with a Catalog UI Policy (show/hide of the existing-ID field, mandatory fields by connection type).
- Order Now flow generating REQ and RITM numbers and a Request Summary page.
- Flow Designer flow that creates auto-numbered DAT... records in u_database_tables.
- Manager approval with the record updated on the decision (approved / else branch).
- Dynamic HTML email notifications (created and approved) logged in sys_email.

3. Architecture

The MERN template's React / Node.js / MongoDB layers map to the ServiceNow platform as follows.

Layer	Equivalent in this project	Details
Frontend	Service Portal (/sp)	Home page, catalog search, the Network Request form (variables: Requested For, Mobile Number, Type of Connection, Existing Id, Total Amount, Mode of Payment, Address, plus Requester Information), Order Confirmation and Request Summary pages. A Catalog UI Policy runs in the browser to show/hide the ID field.
Backend	Flow Designer / Workflow Studio flow 'Network Request'	Triggered by the Service Catalog. Actions: 1 Get Catalog Variables, 2 Create Database Tables Record, 3 Send Email, 4 Ask For Approval on Database Tables, 5 Update record (If Request is Approved), 6 Update record (Else).
Database	Custom table u_database_tables (Database Tables)	Fields include Database Number (DAT... auto-number), Mobile Number, Type of Connection, Mode of Payment, Total Amount, Customer Address and Assignment to. Emails are stored in the sys_email table.

Automated Network Request Management System - Solution Architecture

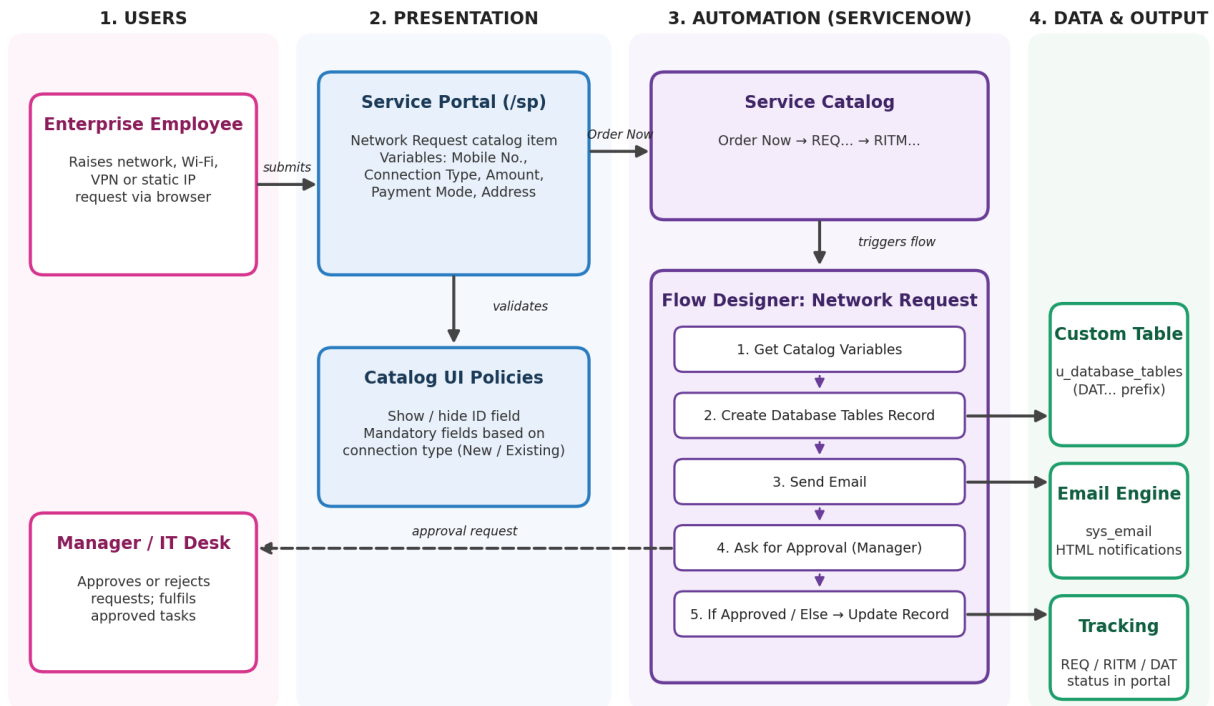


Figure 1: Solution architecture

4. Setup Instructions

Prerequisites:

- A ServiceNow instance (a Personal Developer Instance was used) and a user with the admin role.
- A desktop web browser (Chrome).
- An email configuration (SMTP) for outbound notifications on the instance.

Installation / configuration steps:

1. Create the custom table: System Definition > Tables > New, label 'Database Tables', name u_database_tables, with the fields listed above and a DAT number prefix.
2. Create the catalog item 'Network Request' (category Network Standard Changes) and add its variables.
3. Create the Variable Set 'Requester Information' (Opened on behalf of, User Name, Email ID, Phone Number) and attach it to the item.
4. Add the Catalog UI Policy 'Hide/Show of ID field' to the catalog item.
5. Build the flow in Workflow Studio: trigger = Service Catalog; add the six actions and map the catalog variables to the table fields.
6. Configure the Send Email action (subject and dynamic body) and the approval conditions, then activate the flow.
7. Open the Service Portal (/sp), search 'network request' and place a test order.

5. Folder Structure

ServiceNow is a low-code platform, so there are no client / server folders. The application artifacts are organised as follows:

Area	Artifacts
Client (frontend)	Service Portal (/sp) pages, catalog item 'Network Request', Variable Set 'Requester Information', Catalog UI Policy 'Hide/Show of ID field'
Server (backend)	Flow 'Network Request' (Workflow Studio), approval conditions, Send Email action, notification content
Data	Table u_database_tables, request tables (REQ / RITM), email log table sys_email

6. Running the Application

There are no npm commands; the application runs on the instance once the flow is active.

- **Frontend:** open <https://dev393606.service-now.com/sp> in the browser and open the Network Request item.
- **Backend:** the Network Request flow starts automatically when an order is placed; its execution can be checked in Workflow Studio.

7. API Documentation

The project does not define custom REST endpoints. Records can still be read or created through the standard ServiceNow Table API (a platform feature), for example:

Method	Endpoint	Purpose
GET	/api/now/table/u_database_tables	List Database Tables records (supports sysparm_query, sysparm_limit)
GET	/api/now/table/u_database_tables/{sys_id}	Get one record
POST	/api/now/table/u_database_tables	Create a record from a JSON body
PATCH	/api/now/table/u_database_tables/{sys_id}	Update fields (for example the approval status)

Example request body and response (field values are illustrative):

```
POST /api/now/table/u_database_tables
{ "u_mobile_number": "9000000000",
  "u_mode_of_payment": "UPI",
  "u_total_amount": "5000",
  "u_customer_address": "Hyderabad" }

Response 201: { "result": { "sys_id": "...",
  "u_database_number": "DAT...", ... } }
```

The column names above follow the table fields seen in the project; confirm the exact column names in the table dictionary before using the API.

8. Authentication

- Users sign in to the instance with their ServiceNow account (session-based login). The Service Portal shows the signed-in user, for example System Administrator in the test run.
- Authorization is role-based: employees order from the catalog, managers receive and decide approvals, administrators configure the flow and table.
- The approval step in the flow enforces manager sign-off before the record is marked approved.
- For the REST Table API, the platform supports basic authentication and OAuth tokens; REST access follows the same role and ACL rules (the table has access controls for read, create, write and delete).

9. User Interface

Key screens of the Service Portal are shown below.

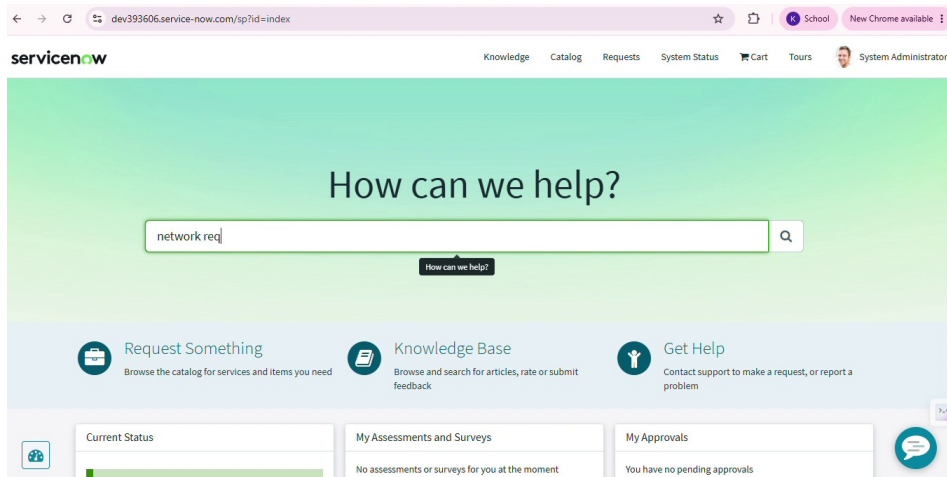


Figure 2: Service Portal home page

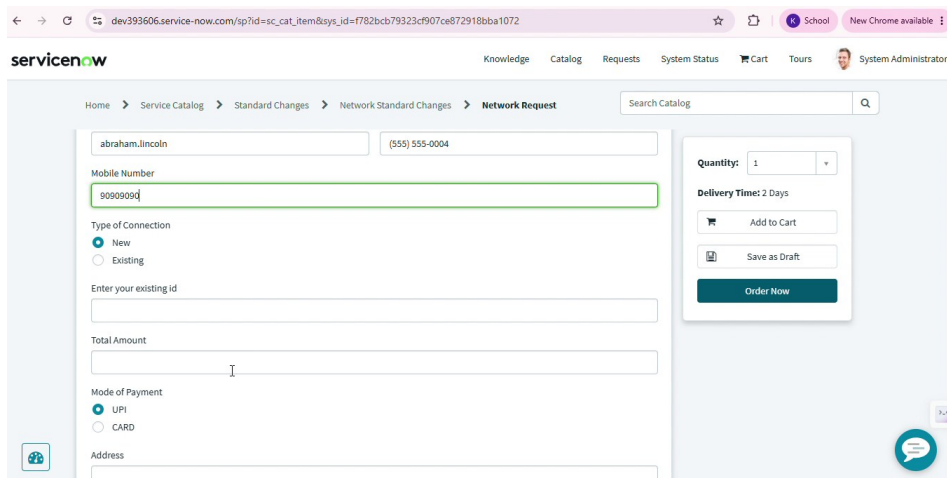


Figure 3: Network Request form

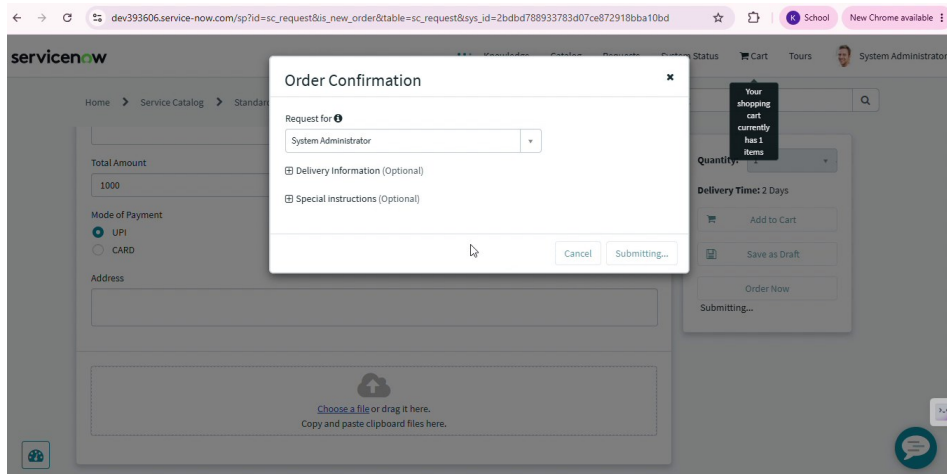


Figure 4: Order Confirmation

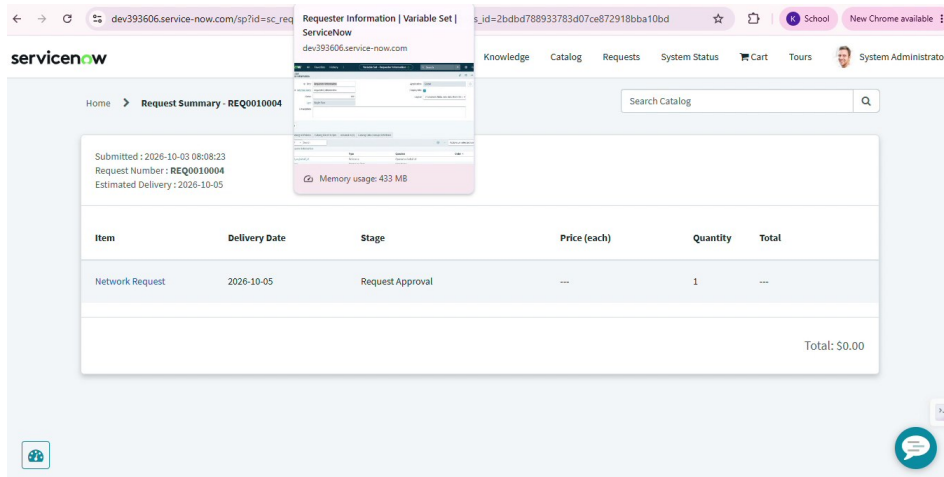


Figure 5: Request Summary

10. Testing

Testing strategy: functional tests of the form, order and flow; performance tests of notification time and multiple requests; and user acceptance testing (UAT). Tools: Service Portal, Workflow Studio flow execution view and the sys_email log.

Area	Result
Functional tests (FT-01 to FT-04)	All 4 passed
Performance tests (PT-01 to PT-03)	All 3 passed; notification generated in about 1 second
UAT (10 test cases)	9 passed, 0 failed, 1 not tested (TC-010)
Requests processed	3 of 3 (REQ0010002, REQ0010003, REQ0010004) with created and approved emails

11. Screenshots or Demo

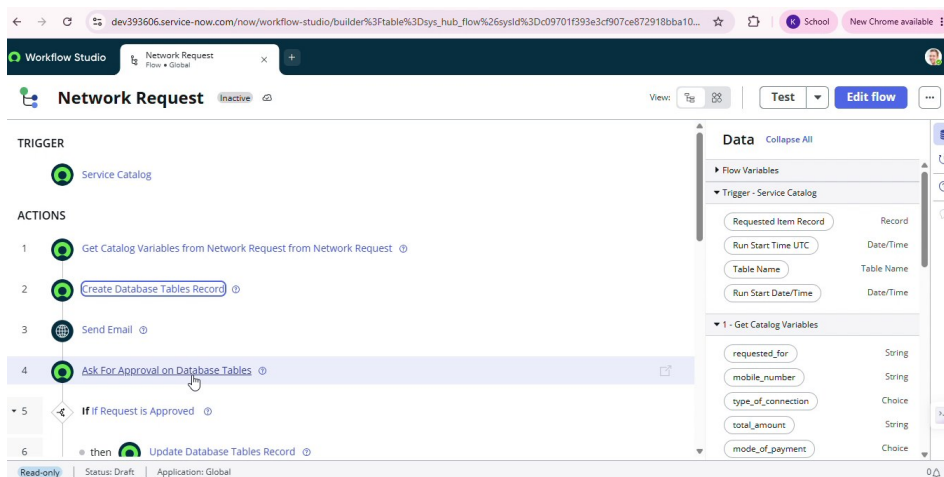


Figure 6: Network Request flow

Created	Recipients	Subject	Type	Notification type	User ID
2026-10-03 02:49:55	fred.luddy@example.com	Catalog Task SCTASK0010004 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 02:55:15	fred.luddy@example.com	Catalog Task SCTASK0010006 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 08:08:25	fred.luddy@example.com	Catalog Task SCTASK0010008 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 00:00:15	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
about 5 hours ago	admin@example.com	Request REQ0010002 was approved	send-ready	SMTP	(empty)
2026-10-03 02:55:13	admin@example.com	Request REQ0010003 was approved	send-ready	SMTP	(empty)
2026-10-03 08:08:25	admin@example.com	Request REQ0010004 was approved	send-ready	SMTP	(empty)
2026-10-03 02:49:54	admin@example.com	Request REQ0010002 was created	send-ready	SMTP	(empty)
2026-10-03 02:55:14	admin@example.com	Request REQ0010003 was created	send-ready	SMTP	(empty)
2026-10-03 08:08:25	admin@example.com	Request REQ0010004 was created	send-ready	SMTP	(empty)
2026-10-03 01:00:05	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-10-03 01:00:06	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-10-03 01:00:07	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)

Figure 7: Emails log (created and approved notifications)

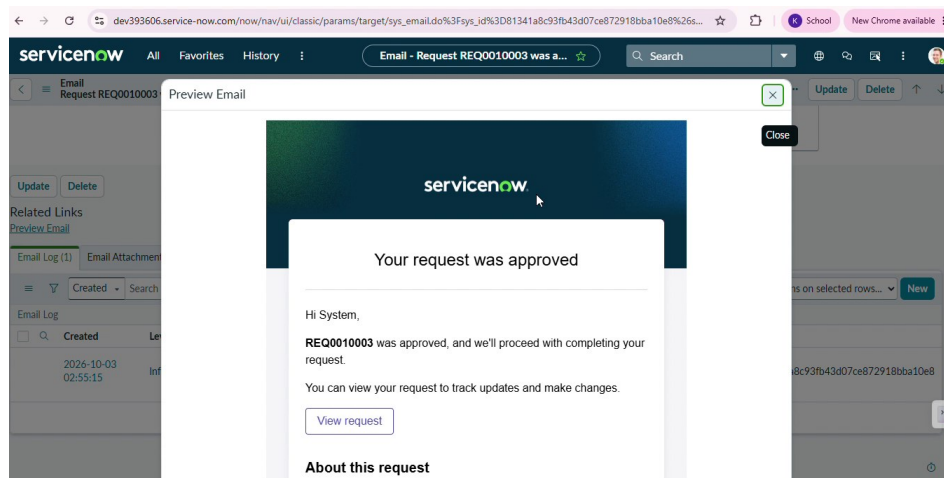


Figure 8: Approval email preview

Demo link: to be added by the team (a screen recording of the project was prepared).

12. Known Issues

- BG-001 (Medium): the 'Enter your existing Id' field is still visible when Type of Connection is New; review the On load and Reverse if false settings of the UI policy.
- BG-002 (Low): notification emails go to sample example.com addresses on the developer instance; configure real recipients and SMTP before production.
- TC-010: the negative test (missing mandatory details) has not been executed yet.

13. Future Enhancements

- SLA timers and escalation for pending approvals.
- Virtual Agent / chatbot for request status.
- Dashboards on request volumes and approval times.
- Integration with network devices and IP address management for automatic provisioning.
- Separate catalog variants for VPN, Wi-Fi and static IP requests.